

NAZIA SULTANA MARJAN

Computer Science & Engineering | AI/ML & NLP Researcher

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SUMMARY

Computer Science and Engineering graduate (CGPA 3.78/4.00) specializing in Artificial Intelligence, Machine Learning, Natural Language Processing, and Large Language Models. Experienced in academic research, AI/ML projects, and scientific publication, with an undergraduate thesis on Bengali healthcare question answering and an IEEE conference paper. Interested in teaching, student mentoring, research, academic publishing, and contributing to academic and departmental activities.

EDUCATION

B.Sc. in Computer Science & Engineering

September 2022 – August 2026

Daffodil International University, Dhaka, Bangladesh | **CGPA: 3.78 / 4.00**

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Machine Learning, Natural Language Processing, Web Engineering, Mobile Application Development

RESEARCH INTERESTS

Artificial Intelligence & Machine Learning | Natural Language Processing | Large Language Models | Generative AI | Retrieval-Augmented Generation | Low-Resource Language AI | Healthcare AI | Multilingual NLP | LLM Fine-Tuning / Model Adaptation

RESEARCH EXPERIENCE

Undergraduate Thesis - HeallM: An Intelligent Medical Assistant for Healthcare Query Management Based on Bengali Data

Daffodil International University

2025–2026

- Researched and developed a Bengali-focused AI healthcare assistant for multilingual medical query management, addressing question answering in a low-resource language.
- Applied LoRA, QLoRA, 4-bit quantization, and parameter-efficient fine-tuning (PEFT) for large language model adaptation.
- Developed a Knowledge Graph and GraphRAG-based retrieval framework to incorporate structured medical knowledge and improve context-aware retrieval and reasoning for healthcare queries.
- Used NLLB-200 to support Bengali English multilingual processing within the system.

PUBLICATION

1. IoT-Enabled Crop Storage System: An Integrated Approach to Reduce Post-Harvest Losses. *IEEE QPAIN 2025*. DOI: 10.1109/QPAIN66474.2025.11171780 (IEEE Xplore).
2. MedImBD: An AI-Driven Framework for Bengali Healthcare Query Management. *IEEE Asia-Pacific Conference on Computer Science and Data Engineering (IEEE CSDE 2026)*. Paper under review.

TEACHING & MENTORING EXPERIENCE

Private Tutor

Bangladesh

- Provided one-to-one and small-group academic instruction (private tutoring).
- Explained concepts according to students' learning levels and prepared lessons and exercises.
- Monitored student progress and provided academic guidance.

TEACHING INTERESTS

Programming Fundamentals | Object-Oriented Programming | Data Structures & Algorithms | Database Management Systems | Operating Systems | Machine Learning | Artificial Intelligence | Natural Language Processing

TECHNICAL SKILLS

Programming: Python, C, C++, SQL

Core CSE: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Web Engineering, Mobile Application Development

AI/ML: Machine Learning, Deep Learning, PyTorch, Scikit-learn

NLP/LLM: NLP, LLMs, RAG, Transformers, Hugging Face, NLLB-200, QLoRA, PEFT, Prompt Engineering

Data: Pandas, NumPy, FAISS, Data Analysis

Development: Streamlit, Flutter, Firebase, REST APIs

Tools: Git, GitHub, Jupyter, Google Colab, VS Code

SELECTED PROJECTS

MedLMDBD: An AI-Driven Framework for Bengali Healthcare Query Management

2025–2026

- Developed a multilingual healthcare framework using SmolLM, MedAlpaca, QLoRA, PEFT, and NLLB-200.
- Integrated ChatDoctor and MedInstruct-52K datasets for model adaptation and evaluation.

- RetinaDetect AI: AI-Powered Retinal Disease Screening System** 2026
- Developed an EfficientNetV2B3-based retinal image classifier for cataract, glaucoma, diabetic retinopathy, and normal cases; built and deployed an interactive Streamlit screening application.
- IoT-Enabled Crop Storage System (basis of IEEE QPAIN 2025 publication)** 2025
- Developed an IoT-based system for monitoring storage conditions and reducing post-harvest losses, implementing sensor-based environmental monitoring and automated control mechanisms.

PROFESSIONAL EXPERIENCE

- AI Intern - Decode Labs, Remote** July 2026 – August 2026
- Contributed to AI-related tasks and applied machine learning knowledge to practical projects in a professional environment.

LEADERSHIP & ACADEMIC ACTIVITIES

- Executive Member - Girls Computer Programming Club (GCPC), DIU** 2023–2025
- Participated in programming activities, technical events, and peer-learning initiatives.
- Volunteer - Daffodil Air & Rover Scout Group** 2023–2025
- Participated in team-based volunteering, organizational, and leadership activities.

ACHIEVEMENTS

- **Academic Merit Scholarship** - Received tuition fee waiver based on academic performance throughout the B.Sc. in CSE program at Daffodil International University.
- **Unlock the Algorithm Programming Contest** -2024.
- **DIU Take Off Programming Contest** - 2022.